# Pedro Lucas Reis

## Full Stack Developer | AI Engineer (LLMs, RAG, MCP)

React, Next.js, TypeScript, Python, C#
Volta Redonda, RJ, Brazil | Remote
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Portfolio | LinkedIn | GitHub | Hugging Face

## Professional summary

I build full-stack applications and integrate LLMs into products with RAG, MCP, tool use, evaluations and fine-tuning. I have worked remotely since 2022. I orchestrated 20 concurrent agents using MCP and published the Argos-Action LoRA adapter on Hugging Face.

## Professional experience

### Domus | Sua Solução Digital

Mid-level Full Stack Software Engineer · Applied AI | Oct/2025 / Present · Remote

- Built RAG pipelines with embeddings, chunking, reranking, vector databases and graph databases.
- Orchestrated Claude, GPT and Gemini workflows using MCP, tool use and function calling. Used structured outputs and Jev to evaluate prompts, catch regressions and trace actions, with controls for prompt injection, cost and latency.
- Built AI and SQL anomaly monitoring in n8n (70% less analysis time), churn prediction (85% accuracy) and fraud detection (75% less triage time).
- Automated collections through WhatsApp, email and SMS, reducing manual work by 60% and improving recovery by 20%. A SQL Server assistant on WhatsApp reduced reporting time by 80%.
- Used Zapier and CRM workflows to cut sales qualification time by 65% and cancellations by 25%. In Make, integrated orders, inventory and invoicing, reducing manual processing by 80% and data-entry errors by 90%.
- Applied AI to SQL and API data analysis, reducing analysis time by 75% and reporting effort by 60%.
- Delivered Propose, a SaaS proposal builder with a drag-and-drop editor, JSON Schema, PostgreSQL, Supabase and payments. Built a Go proxy for routing, caching and retries.

### Autocom3

Mid-level Full Stack Software Engineer · Applied AI | Nov/2024 / Present · Hybrid

- Lead a React and TypeScript design system with more than 500 components. Cut delivery time for new screens by up to 70%, from about three days to a few hours.
- Built the unified portal in Next.js with more than 120 pages, along with accounting, customer and administrative portals.
- Used TanStack Query to reduce redundant network traffic by about 60% and improve load times by up to 40%, with Time to Interactive under 1.2 s.
- Orchestrated 20 concurrent agents using MCP and tool use. Evaluated structured outputs with Jev and used a graph database, with controls for cost, latency and prompt injection.
- Built APIs in Node.js, NestJS and C# with JWT, RBAC, rate limiting and webhooks. Modeled data in PostgreSQL and SQL Server and deployed to AWS via API Gateway.
- Helped migrate Electron apps to Tauri, added Playwright E2E tests to AC3 Cloud and maintained customer-facing Vue.js applications.

## Earlier experience and mentoring

### EVA.Tech · UniFOA / Sebrae

Technology & AI Mentor | Jun/2024 / Dec/2024

- Mentored more than 30 participants in programming logic, web development and applied AI over six months.
- Led hands-on ideation and prototyping workshops.

### SIVIS Tecnologia

Junior Full Stack Engineer | Aug/2022 / Jun/2024

- Built operational dashboards, financial workflows, and real-time analytical reporting.
- Third-party REST API integrations (payment processors, registry services, messaging).

## Technical skills

**AI and automation.** Python, PyTorch, LLMs, RAG, embeddings, vector and graph databases, chunking, reranking, MCP, tool use, function calling, prompt engineering, structured outputs, Jev evals, prompt injection defenses, LoRA, Hugging Face, n8n, Zapier, Make.

**Frontend.** TypeScript, React, Next.js (SSR and SSG), Vue.js, Angular, Tailwind CSS, design systems and TanStack Query.

**Backend and data.** Node.js, NestJS, C#, Express, Fastify, Go, REST APIs, JWT, RBAC, webhooks, PostgreSQL, SQL Server, MySQL, Supabase and AWS API Gateway.

**Quality and desktop.** Playwright, Electron, Tauri and Git.

## Projects and publications

### Argos-Action

Open Computer Use prototype using Qwen2.5-1.5B-Instruct, LoRA and accessibility trees. In 132 new cases, 92% of responses matched the schema, compared with 0% for the base model. None of the 28 critical cases were escalated. GitHub | Technical article

### Achilles CDP Agent

Personal open-source Python MCP server for controlled browser and Windows automation, with handoff to a human operator. GitHub

### Organon

Private personal project. Local-first desktop cockpit in the Reoli suite for sessions across models and human approvals.

### FalaAtípica

Capstone on assistive technology that helps families and therapists support children with speech delays. UniFOA proceedings

## Education and languages

Bachelor's degree in Information Systems - UniFOA. Completed December 13, 2025. English B2.